Today's Session

Part 1: Orientation 2:00–3:00 PM

- Course overview, structure & rules
- The Big Data landscape in 2026
- Our complete tool stack
- 17-week journey overview
- Assessment, project & expectations

Part 2: Hands-On Setup 3:10–4:30 PM

- Hardware & software check
- Docker Desktop installation
- Launching the Big Data stack
- First PySpark job (live coding)
- Exploring the Spark Web UI

Take-Home Assignment

Due Sunday 15 March, 23:59

Complete full environment setup + run the getting-started notebook + submit screenshots and a 100-word reflection on Google Classroom.

Course Snapshot

Course No.	Z5008
Title	Big Data Lab
Credits	6 (3 hrs contact + 6–9 hrs self-work/week)
Slot	Monday 2:00–4:40 PM (Slot P)
For	MTech DS&AI - Semester 2 (Core)

Prerequisites:

- Python programming (intermediate level)
- Basic data structures & databases
- Laptop with $\geq$8 GB RAM & 20 GB free disk

Assessment at a Glance

Written Quiz (Week 7)	15%
Mid-Sem Lab Exam (Week 13)	15%
End-Sem Lab Exam	20%
Course Project	**50%**
Total	**100%**

The **project** is 50% of your grade. Start planning your domain from **Day 1**. Individual.

Why Big Data Matters in 2026

The Scale of the Problem

- **4.6 quintillion bytes** generated every single day
- **75+ billion** IoT devices active worldwide
- LLMs trained on **petabytes** of curated text
- Real-time AI demands **sub-second** decisions
- 90% of all data created **in the last 2 years**

The Career Opportunity

- Data Engineers earn are well paid
- 3 million+ open data roles globally (2025)
- Every company is now a data company

Real-World Applications

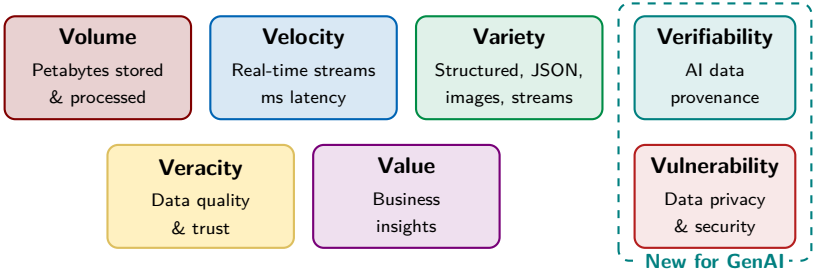
- **Healthcare:** Genomic pipelines, clinical AI
- **Agriculture:** Precision farming, sensors
- **Finance:** Fraud detection ($<$50 ms latency)
- **Retail:** Real-time personalisation
- **Transport:** Fleet & logistics optimisation
- **Climate:** Petabyte-scale weather models

The African Context

Mobile money, precision agriculture, health data pipelines - Big Data is **Africa's next competitive frontier**.

This is your edge.

The 5 V's of Big Data (+ 2 New Ones for 2026)



Why Verifiability & Vulnerability?

In the LLM era, knowing *where* your data came from (Provenance) and ensuring it doesn't leak personal information (Privacy) are no longer afterthoughts, they dictate your entire pipeline architecture.

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The Modern Big Data Stack (2026)

SERVING & MONITORING
BentoML · Grafana · Prometheus · REST APIs

ML & MLOps
Spark MLlib · Ray · MLflow · Great Expectations

PROCESSING & ORCHESTRATION
Apache Spark 3.5+ · Airflow · Dagster · dbt

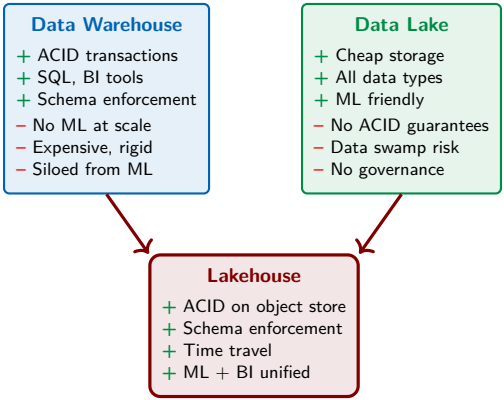
LAKEHOUSE FORMAT
Delta Lake · Apache Iceberg · Schema Evolution · Time Travel

OBJECT STORAGE
MinIO (S3-compatible) · Parquet · ORC

INGESTION
Kafka 3.7+ · Redpanda · REST APIs · CDC

We will build proficiency at every layer of this stack over our 17-week journey.

The Lakehouse Paradigm



Key Technologies

- **Delta Lake** — Databricks
- **Apache Iceberg** — Netflix/Apple
- **Apache Hudi** — Uber

This Course

We use **Delta Lake** and **Iceberg** on **MinIO** (S3-compatible local storage) throughout the semester.

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Your Technology Stack for This Course

Category	Tool(s)
Language	Python 3.11+ (primary)
Big Data Engine	Apache Spark 3.5+
Lakehouse Format	Delta Lake, Iceberg
Object Storage	MinIO (S3-compatible)
Streaming	Apache Kafka 3.7+
Orchestration	Airflow 2.9, Dagster
ML / Features	Spark MLlib, Ray
MLOps	MLflow, Great Expectations
Deployment	BentoML, Docker
Monitoring	Grafana, Prometheus
IDE	JupyterLab, VS Code
Cloud (backup)	Databricks Community Ed.

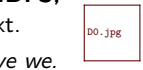
Why These Tools?

- **Industry standard** — Netflix, Spotify, Grab
- **Open source** — free forever, career-portable
- **2026 relevant** — Lakehouse + MLOps is the market
- **Docker-based** — run the stack locally
- **Cloud-portable** — same code on AWS/GCP/Azure

Deprecated Context

We will only briefly cover **Hadoop, HDFS, and MapReduce** for historical context.

The industry has moved on — so have we.



Your 17-Week Journey

FOUNDATIONS	EXAMS	STREAMING	ML & MLOps	PROJECT
Weeks 1–7	W. 8	W. 9–10	Weeks 11–15	W. 16–17

W	Topic
1	Course orientation + environment setup
2	Lakehouse architecture + Delta Lake intro
3	Delta Lake advanced + Apache Iceberg
4	Spark 3.5+ foundations & DataFrame API
5	Spark SQL + joins + window functions
6	Spark performance tuning + AQE
7	Orchestration: Airflow + Dagster
8	**MID-SEMESTER: Written Quiz + Lab**

W	Topic
9	Real-time streaming: Kafka + Redpanda
10	Spark Structured Streaming + stateful
11	Visualisation: Grafana + Streamlit
12	Distributed ML: MLlib + Ray
13	MLOps: MLflow tracking + registry
14	Model deployment: BentoML + Docker
15	Monitoring + scaling + data governance
16	Project integration + load testing
17	**FINAL PROJECT PRESENTATIONS**

Assessment in Detail

Grading Distribution

Component	Weight
Written Quiz (Week 8)	15%
Mid-Sem Lab Exam (Week 13)	15%
End-Sem Lab Exam (W. 17+)	20%
Course Project	**50%**
Total	**100%**

Project (50%) Breakdown

Deliverable	Weight
Proposal + Architecture	10%
Progress Reviews ($\times 2$)	10%
GitHub Repo Quality	10%
Final Demo + Load Test	10%
Presentation + 8-min Video	10%

Lab Exam Format

- Individual work; open-notes allowed
- Pre-configured Docker environment
- Tasks: Write Spark jobs, debug pipelines, query tables
- Graded on: Correctness, efficiency, style

Academic Integrity

AI coding tools (Copilot, ChatGPT) are **allowed** but **must be declared**. You must be able to explain every line of code during Q&A. Identical submissions will receive a zero.

The Course Project: What You Will Build

Real-Time Lakehouse + MLOps Intelligence Platform

Build a production-grade, end-to-end data system **individually or in pairs (max 2)**.
Choose any domain: agriculture, health, finance, transport, environment, retail...

Required Components:

1. Streaming ingestion pipeline (Kafka)
2. Lakehouse storage layer (Delta Lake/Iceberg)
3. Batch + streaming processing (Spark)
4. Orchestrated workflows (Airflow/Dagster)
5. ML model trained at scale (MLlib/Ray)
6. MLflow: experiment tracking + registry
7. Deployed model API (BentoML + Docker)
8. Live monitoring dashboard (Grafana)

Milestone Timeline:

- **Week 2:** Project brief released
- **Week 3:** Partner selection & domain choice
- **Week 4:** Proposal + architecture diagram
- **Week 8:** Progress Review 1 (10m demo)
- **Week 12:** Progress Review 2 (arch. check)
- **Week 15:** Code freeze
- **Week 16:** Load test + dress rehearsal
- **Week 17:** Final presentation + live demo

Bonus (+5%): Add a natural-language query interface using an LLM e.g., GPT-4o, Gemini, local Llama over your data pipeline.

Lab Policies & Expectations

Attendance & Participation

- $\geq$85% attendance is **mandatory**
- Missing a lab = missed irreplaceable practice
- Arrive on time; >15 min late = absent
- **Bring your charged laptop every session**

During Lab Sessions

- Focused work; no social media/gaming
- **Collaboration encouraged** - help neighbours
- Ask questions loudly - no silly questions
- Document your errors; they are your best teacher

Weekly Assignments

- Not graded - but **exam questions come from them**
- Complete *before* the next session
- Discussed at the start of each lab (15 min)
- Submit on Moodle for feedback

Communication

- Email: `innocent@iitmz.ac.in`
- Response time: within 24 h on weekdays
- Google Classroom: for urgent setup questions
- Office hours: by appointment (email first)
- Stack Overflow & docs: **use them first**

10-Minute Break

Back at **3:10 PM** for hands-on setup

Stretch, hydrate, use the bathroom

"In God we trust. All others must bring data."

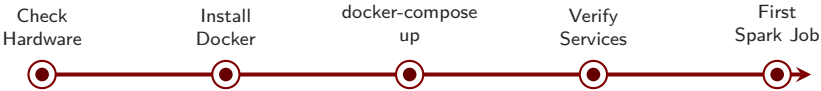
— W. Edwards Deming

Our Goal for the Next 80 Minutes

Have a fully working Big Data development environment on your laptop:

Apache Spark 3.5 · **Delta Lake** · **Apache Kafka** · **MinIO** · **MLflow**

all running locally via a single `docker-compose up` command.



See the printed `setup_guide.pdf` handout in Google Classroom.

Hardware & Software Requirements

Hardware Requirements

- CPU: 64-bit (Intel/AMD/Apple Silicon)
- RAM: **8 GB minimum** (16 GB rec.)
- Disk: **20 GB free** (for Docker images)
- OS: Windows 10/11, macOS 12+, Ubuntu

Apple Silicon (M1/M2/M3)?

Docker Desktop supports ARM64. All images in our stack have ARM64 variants. If an older image fails, you may need:

```
platform: linux/amd64
```

Software to Install

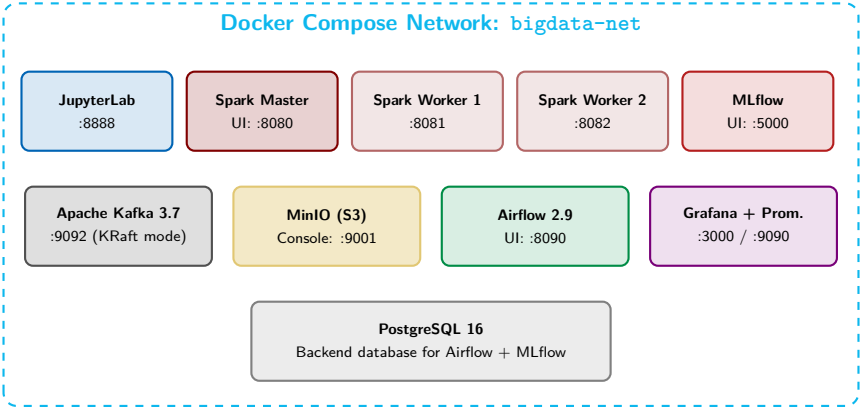
1. **Docker Desktop** (free)
   `docker.com/products/docker-desktop`
2. **VS Code** + Python extension
   `code.visualstudio.com`
3. **Git** (for course materials)
   `git-scm.com`

Laptop Struggling? Backup Options:

Use **Databricks Community Edition** (free cloud): `community.cloud.databricks.com`. Sign up with your `iitmz.ac.in` email.

Google Colab also supports PySpark!

Our Docker Compose Big Data Stack



Warning: Total RAM usage is ≈6–8 GB when all services are running.
Close other heavy applications (like Chrome) during the lab.

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Step 1: Launch the Stack

Open Git Bash (Windows) / Terminal (Mac/Linux):

```
# Create your working directory
mkdir ~/bigdata-lab && cd ~/bigdata-lab

# Copy docker-compose.yml from USB or download from Moodle
# (place it inside ~/bigdata-lab/)

# Copy the environment file
cp .env.example .env

# FIRST-TIME: Pull all Docker images (~8 GB - use USB if WiFi is slow)
docker-compose pull

# Start all services in background (-d = detached mode)
docker-compose up -d

# Verify all containers are running
docker-compose ps
# You should see: State = "running" for all 9 services
```

Using the USB Drive (Faster / No Internet Needed)

Load the pre-cached images directly from the provided USB drive to skip the 8 GB download:

```
docker load -i /path/to/usb/bigdata-images.tar.gz
```

Step 2: Verify All 8 Services

Service	URL	Login	What to Check
JupyterLab	`localhost:8888`	token: `bigdata`	Open a notebook
Spark Master UI	`localhost:8080`	none	2 workers listed
MLflow UI	`localhost:5000`	none	Experiments page
MinIO Console	`localhost:9001`	admin / bigdata123	Dashboard loads
Airflow	`localhost:8090`	admin / admin	DAGs page
Grafana	`localhost:3000`	admin / admin	Home page
Prometheus	`localhost:9090`	none	Targets page
Kafka	`localhost:9092`	—	Test via Python

Something Failed? Quick Debug Steps

1. Check logs: `docker-compose logs <service>`
2. Check RAM: Docker Desktop → 10 GB+
3. Restart: `docker-compose restart`
4. Ask in the class Google Classroom

Your First PySpark Job

Go to `localhost:8888`, open `lab01_getting_started.ipynb`:

```python
from pyspark.sql import SparkSession

# Create SparkSession connected to our cluster
spark = SparkSession.builder \
    .appName("Lab01 - Hello Big Data") \
    .master("spark://spark-master:7077") \
    .config("spark.sql.extensions",
            "io.delta.sql.DeltaSparkSessionExtension") \
    .config("spark.sql.catalog.spark_catalog",
            "org.apache.spark.sql.delta.catalog.DeltaCatalog") \
    .getOrCreate()

print(f"Spark version: {spark.version}")    # Should print: 3.5.x
print(f"Worker nodes: {spark.sparkContext.defaultParallelism}")

# Create your first DataFrame
data = [("Alice", 28, "Nairobi"), ("Bob", 32, "Dar es Salaam"),
        ("Chloe", 25, "Zanzibar"), ("David", 29, "Kampala")]

df = spark.createDataFrame(data, ["name", "age", "city"])
df.show()
df.printSchema()
```

Write Your First Delta Lake Table

```python
import os
# Configure MinIO S3 credentials (already set in docker-compose)
# spark._jsc.hadoopConfiguration().set("fs.s3a.endpoint", "http://minio:9000")

# Write DataFrame as a Delta Lake table to MinIO
df.write \
    .format("delta") \
    .mode("overwrite") \
    .save("s3a://warehouse/lab01/people")

print("Successfully written to Delta Lake on MinIO!")

# Read it back
df_delta = spark.read.format("delta").load("s3a://warehouse/lab01/people")
df_delta.show()

# Explore the Delta transaction log
from delta.tables import DeltaTable
dt = DeltaTable.forPath(spark, "s3a://warehouse/lab01/people")
dt.history().show(truncate=False)   # See the full change history!
```

What Just Happened?

You wrote data to an **object store** (MinIO) using the **Delta Lake** format. This is the **core Lakehouse pattern** we will use all semester. Check the MinIO Console (:9001) to visually inspect the

Explore the Spark Web UI

Open `localhost:8080` in your browser:

- **Jobs:** Every DataFrame action becomes a job
- **Stages:** Jobs split into parallel stages
- **Tasks:** Stages split into tasks per partition
- **Storage:** Cached DataFrames/RDDs
- **Executors:** Worker resources and usage
- **SQL/DataFrame:** Visual query plan (Catalyst)
- **Environment:** Spark configuration

Pro Tip:
The Spark UI is your most powerful debugging tool. We dedicate an entire session to reading it deeply in **Week 6 (Performance Tuning)**. Start getting familiar with it now.

Spark UI: What to Look For

- **Task duration** – shuffle-heavy?
- **GC time** – memory pressure?
- **Data spills** – partition too large?
- **Skew** – one task much slower?
- **Stage DAG** – bottlenecks visible

Try It Now

After running your notebook, go to the **SQL/DataFrame** tab and click on the query.

You will see the visual Catalyst query plan!

Common Issues & Fixes

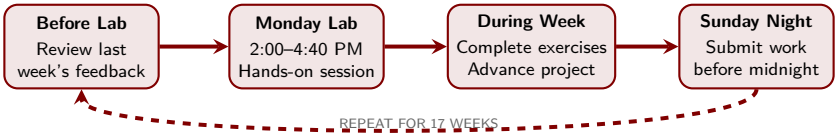
Problem	Solution
`docker pull` fails	Use USB: `docker load -i bigdata-images.tar.gz`
Spark UI: 0 workers	Wait 60 s; check `docker logs spark-worker-1`
JupyterLab blank page	Go to `localhost:8888/lab`, token: `bigdata`
MinIO connection refused	Check `docker-compose ps minio` is *running*
Out of memory (OOM)	Increase Docker RAM to 10 GB (Resources → RAM)
Windows: Docker not found	Ensure WSL2 + Docker Desktop are active
`OutOfMemoryError`	Restart: `docker-compose restart spark-master`
S3 / MinIO "Access Denied"	Check `.env` for `AWS_ACCESS_KEY_ID=admin`

The Reset Option:

```
docker-compose down -v && docker-compose up -d
```

This destroys all containers and volumes (data) to start completely fresh. Use this only if a service is hopelessly corrupted or stuck.

Your Weekly Lab Workflow



Suggested Weekly Time Budget

Activity	Hours	Activity	Hours
Monday Lab Session	3 hrs	Project (Group Work)	2–3 hrs
Notebook Assignments	3–4 hrs	Reading / Videos	1–2 hrs

Total: 10–12 hours per week

Consistency is key in Big Data. Falling behind by two weeks makes catching up very difficult.

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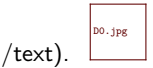
Week 1 Assignment

Due: Sun 15 March, 23:59

Assignment 1: Environment Setup + Exploration

1. **Complete Full Setup:** Submit a screenshot of `docker-compose ps` showing all 9 services in *running* state.
2. **Execute Lab Notebook:** Run all cells in `lab01_getting_started.ipynb`. Add 3 custom DataFrame operations e.g., a new filter or sort.
3. **Exploration Task:** Load any small CSV e.g., from Kaggle:
   - (a) Show basic statistics (`df.describe().show()`)
   - (b) Filter rows by at least one specific condition
   - (c) Group by a column and compute an aggregate (count/avg)
   - (d) Write the final result in **Delta Lake** format to MinIO
4. **Reflection (100 words):** What was the most challenging part of the setup? What are you most excited to learn?

Submission (Google Classroom): Upload `.ipynb` file + `.png` screenshot + Reflection (`.pdf`/text).



Coming Up: Lab 2 - Lakehouse Architecture

Monday 16 March:

- **Deep Dive:** Warehouse vs. Lake vs. Lakehouse
- **Delta Internals:** Transaction logs & checkpointing
- **ACID Operations** on object storage (S3/MinIO)
- **Schema Control:** Enforcement vs. Evolution
- **Hands-on:** MERGE/UPSERT and "Time Travel"

Prepare for Lab 2

- **Read:** *"What is a Lakehouse?"* – Databricks Blog (Link on Moodle)
- **Watch:** *Lakehouse Intro* (15 min)
- **Check:** Ensure your **MinIO** is running!

Project Brief Preview

Released on Google Classroom after Lab 2. Start thinking about your domain: *Agriculture, Health, Finance, or Transport?*

See you in the Lab at 2:00 PM!

Self-Study Expectations (Weekly)

This course cannot be completed from slides alone

- This is the first lecture: these slides are **my summaries**, not the full material.
- We meet only **once per week**, so you must do **significant reading and practice on your own**.
- Use the official **documentation, tutorials, and reference guides** for every tool we touch (Spark, Hadoop, Kafka, Airflow, Docker, etc.).
- Come to lab with questions from your reading and issues you hit while practicing.

Resources: Tools & Recommended Reading

Official docs (start here)

- Apache Spark Documentation
- Apache Kafka Documentation
- Apache Airflow Documentation
- Docker Documentation
- Apache Hadoop Documentation
- Delta Lake (delta.io)
- Apache Iceberg
- MLflow Documentation
- Grafana Documentation

Recommended books / long reads

- Martin Kleppmann: *Designing Data-Intensive Applications*
- Bill Chambers & Matei Zaharia: *Spark: The Definitive Guide*
- Holden Karau et al.: *Learning Spark (2nd ed.)*
- Tyler Akidau et al.: *Streaming Systems*
- Neha Narkhede et al.: *Kafka: The Definitive Guide*

Weekly habit

Read the official docs **while** you build; keep notes of errors, commands, configs, and "gotchas".

Questions?

"The goal of Big Data is to turn data into information, and information into insight."

— Carly Fiorina